

Ore Mineral: A mineral from which one or more metals or metallic compounds can be extracted economically is known as ore mineral.

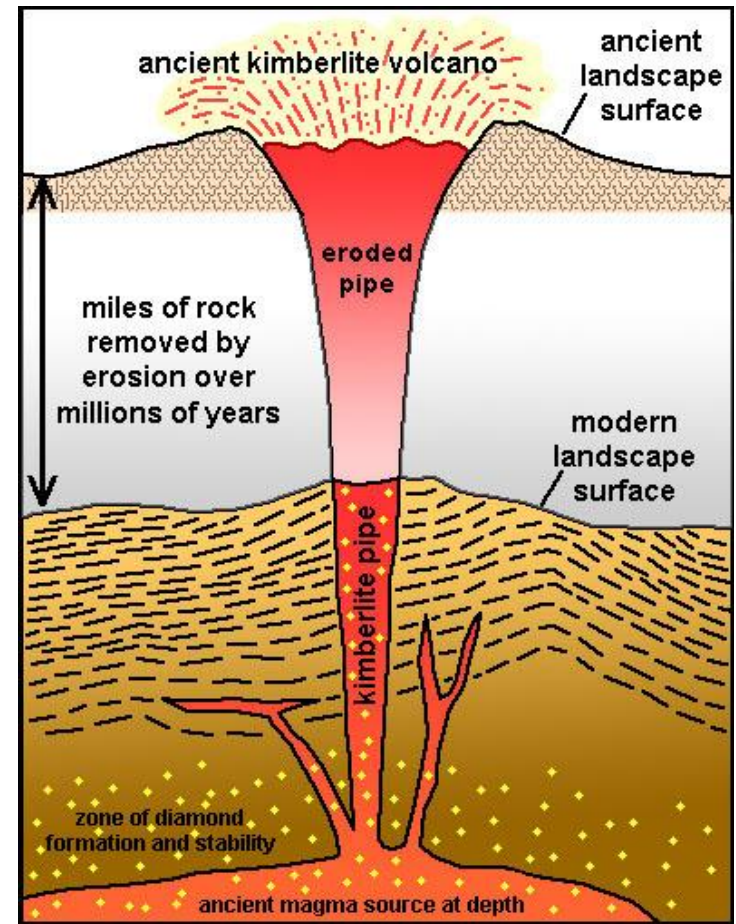
Eg. Chalcopyrite, hematite, galena, sphalerite etc.

Gangue minerals: Gangue minerals include any minerals in an ore deposit that have no economic value. Gangue minerals are normally non-metallic minerals. The gangue minerals are principally silicates and carbonates.

Common gangue minerals:

- Oxides- quartz, hematite etc
- Carbonates- calcite, dolomite
- Sulphates- Barite, Gypsum
- Silicates- Feldspar, clay minerals
- **Miscellaneous- Fluorite, pyrite, pyrrhotite etc.**

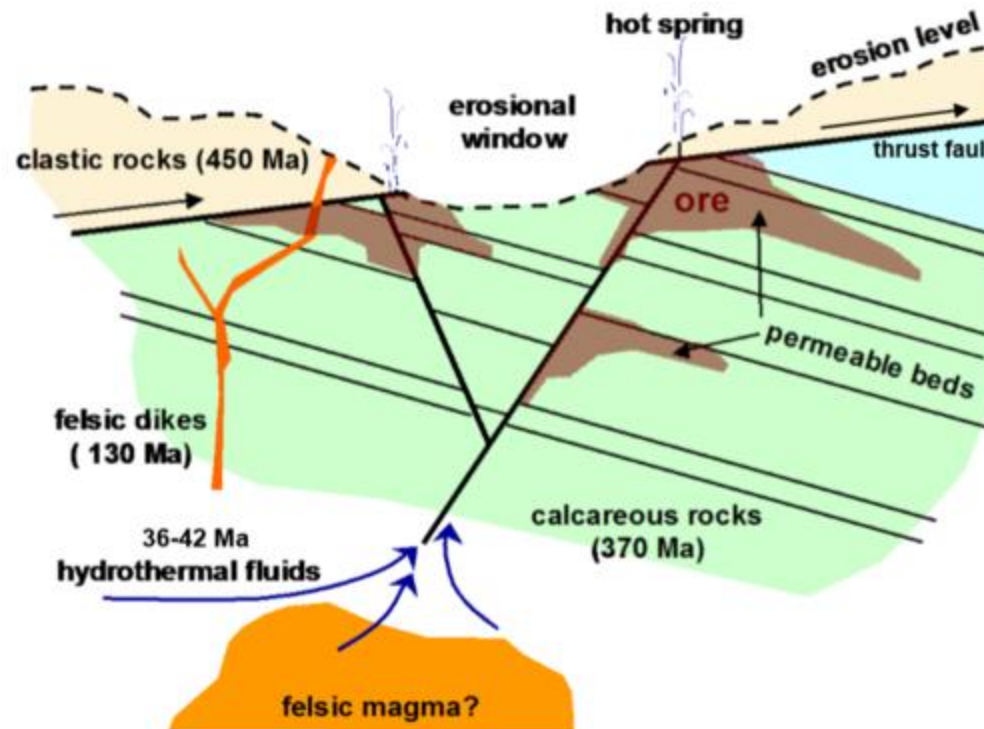
Syngenetic-A syngenetic mineral deposit is a deposit that is formed at the same time as the rocks that enclose it. Magmatic deposits are syngenetic in that the ore minerals crystallize from the same liquid that produces the silicate minerals which form the bulk of the intrusive – they crystallize more or less simultaneously as the melt cools.



Epigenetic-If a mineral deposit formed much later than the rocks which enclose it, it is said to be epigenetic.

An example is a vein.

The first step in the formation of a vein is the fracturing or breaking of rock along a fault zone, at a depth ranging from surface to several kilometers below surface.



Classification of Ore Deposits

Mineral deposits can be classified on the basis of the mechanism responsible for concentrating the valuable substance. Examples Include:

- 1) ***Magmatic Mineral Deposits***
- 2) ***Hydrothermal Mineral Deposits***
- 3) ***Sedimentary Mineral Deposits***
- 4) ***Placer Mineral Deposits***
- 5) ***Residual Mineral Deposits***

Ore Deposit Environments

- **Magmatic**

- **Cumulate deposits** – fractional crystallization processes can concentrate metals (Cr, Fe, Pt)
- **Pegmatites** – late staged crystallization forms pegmatites and many residual elements are concentrated (Li, Ce, Be, Sn, and U)

- **Hydrothermal**

- **Magmatic fluid** - directly associated with magma
- **Porphyries** - Hot water heated by pluton
- **Skarn** – hot water associated with contact metamorphisms
- **Exhalatives** – hot water flowing to surface
- **Epigenetic** – hot water not directly associated with pluton

Ore Deposit Environments

- Sedimentary

- Placer – weathering of primary mineralization and transport by streams (Gold, diamonds, other)
- Banded Iron Formations – 90%+ of world's iron tied up in these.
- Evaporite deposits – minerals like gypsum, halite deposited this way
- Laterites – leaching of rock leaves residual materials behind (Al, Ni, Fe)
- Supergene – reworking of primary ore deposits remobilizes metals (often over short distances)

Magmatic Mineral Deposits

Minerals are concentrated within a body of igneous rock by magmatic processes like crystal settling.

Magmatic process such as partial melting, fractional crystallization, and crystal settling in a magma chamber can concentrate ore minerals containing valuable metals.

As minerals crystallize from a magma body, heavy minerals may sink to the bottom of the magma chamber

1.Early Magmatic:

1.Dissemination: Disseminated crystallization without concentration: eg. Diamond.

2.Segregation: Crystallisation, differentiation and accumulation: eg. Chromite, Singhbhum dt., Jharkhand.

2.Late Magmatic:

1.Gravitative liquid accumulation

•**Residual liquid segregation:** Crystallisation, differentiation and residual magma accumulation: eg. Chromites&Vanadiferous, Titanomagnetite of Mayurbhanj Dt. Orissa.

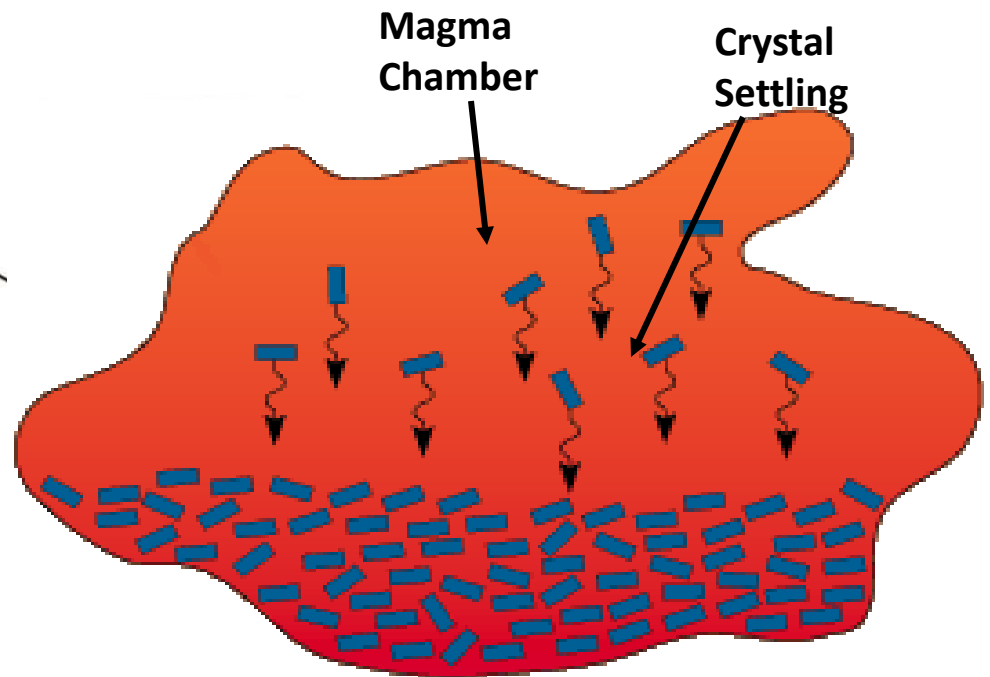
ii. **Residual liquid injection:** Crystallisation, differentiation. Residual magma accumulation with filter pressing and/or injection: eg. Pegmatites hosted mica deposits, Giridih, Jharkhand.

1.Immiscible Liquid accumulation

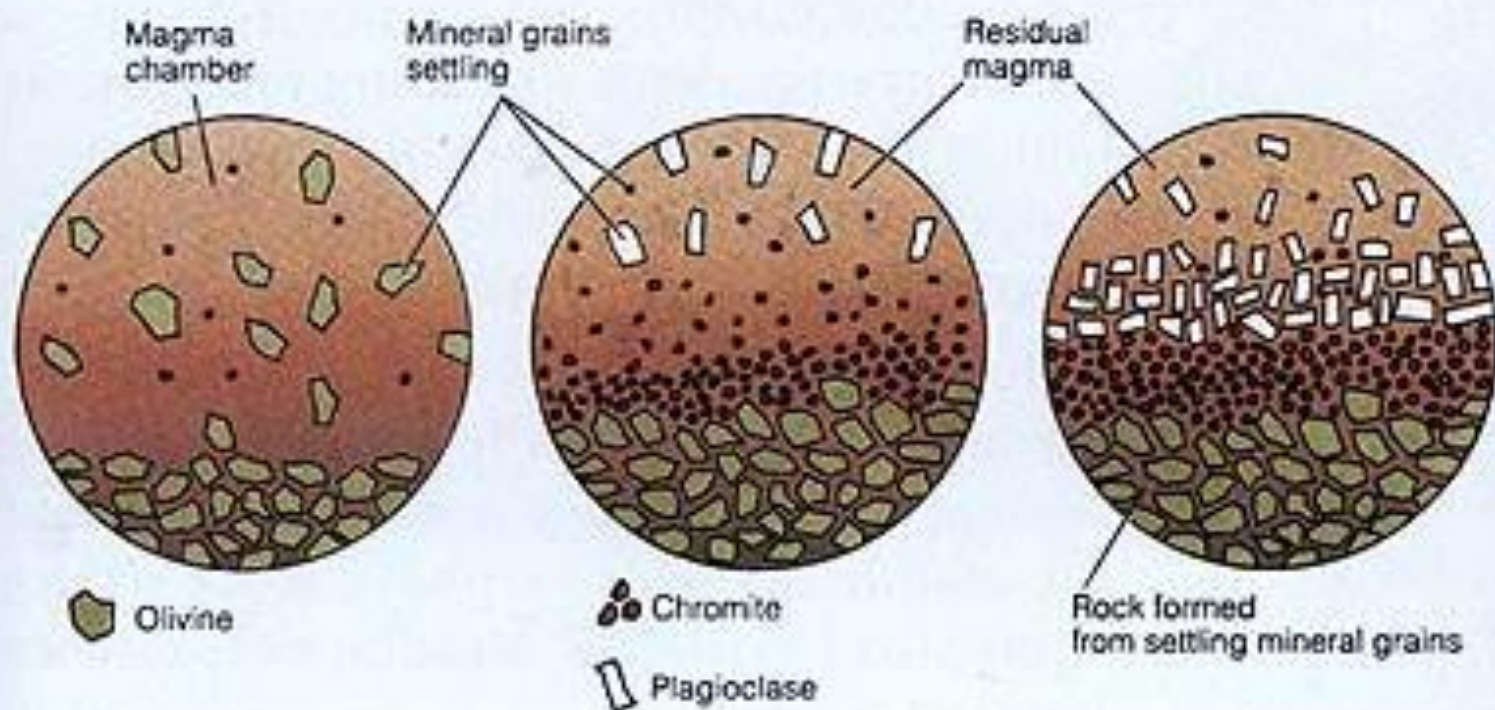
Immiscible liquid segregation: Immiscible liquid segregation and accumulation: eg. 1. Magmatic ilmenite deposits ; 2. Ni-Sulphide deposits.

1. Magmatic Segregation Deposits – General Setting

- Important ore-forming processes all occur within magma chambers
- Minerals accumulate on the floor of magma chambers
- Formed from basic or ultrabasic magma (<52% silica) of mantle origin
- Deposits form by:-
 - Accumulation of crystals
 - Separation of immiscible crystals

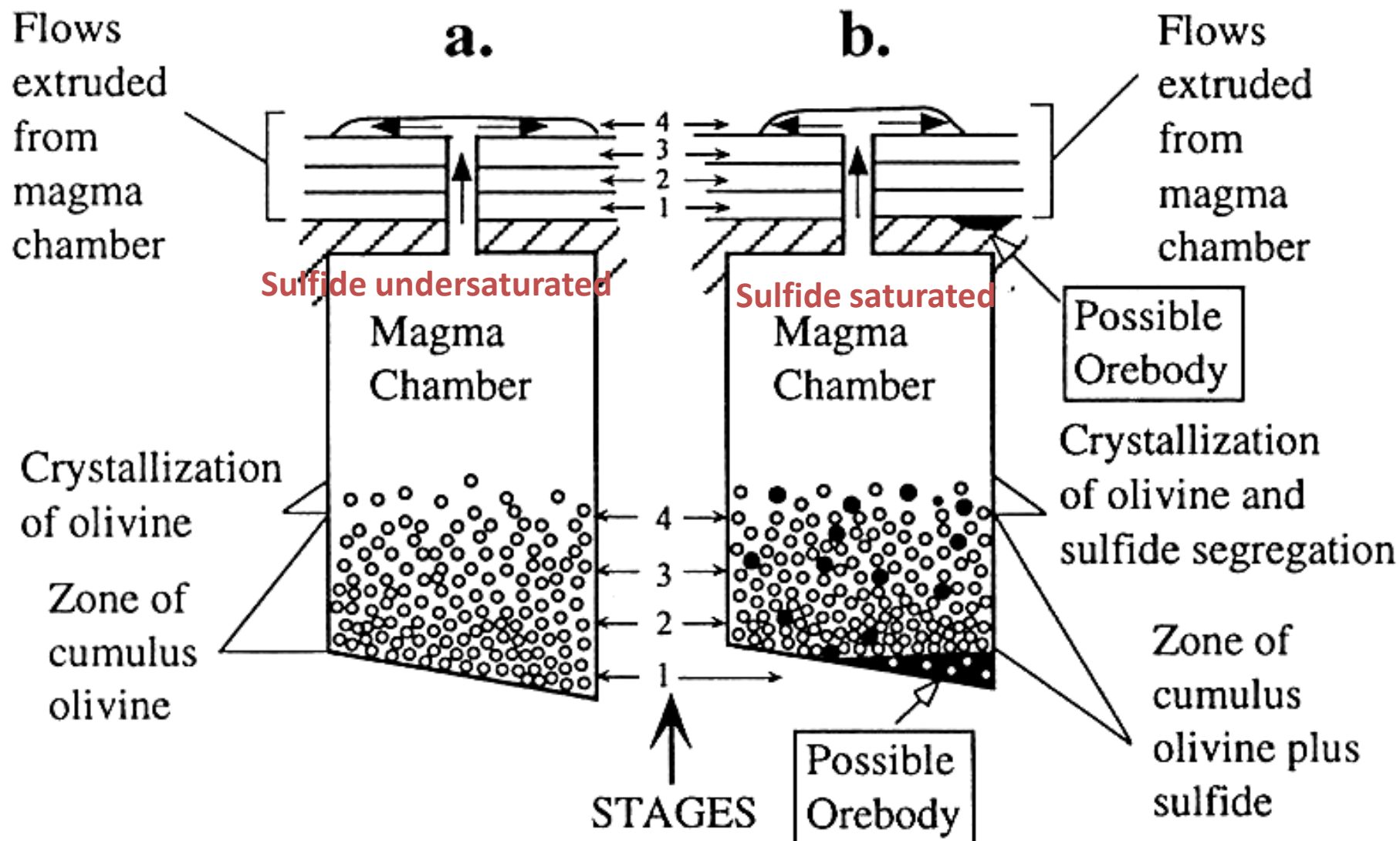


Formation of magmatic mineral deposit by crystal settling

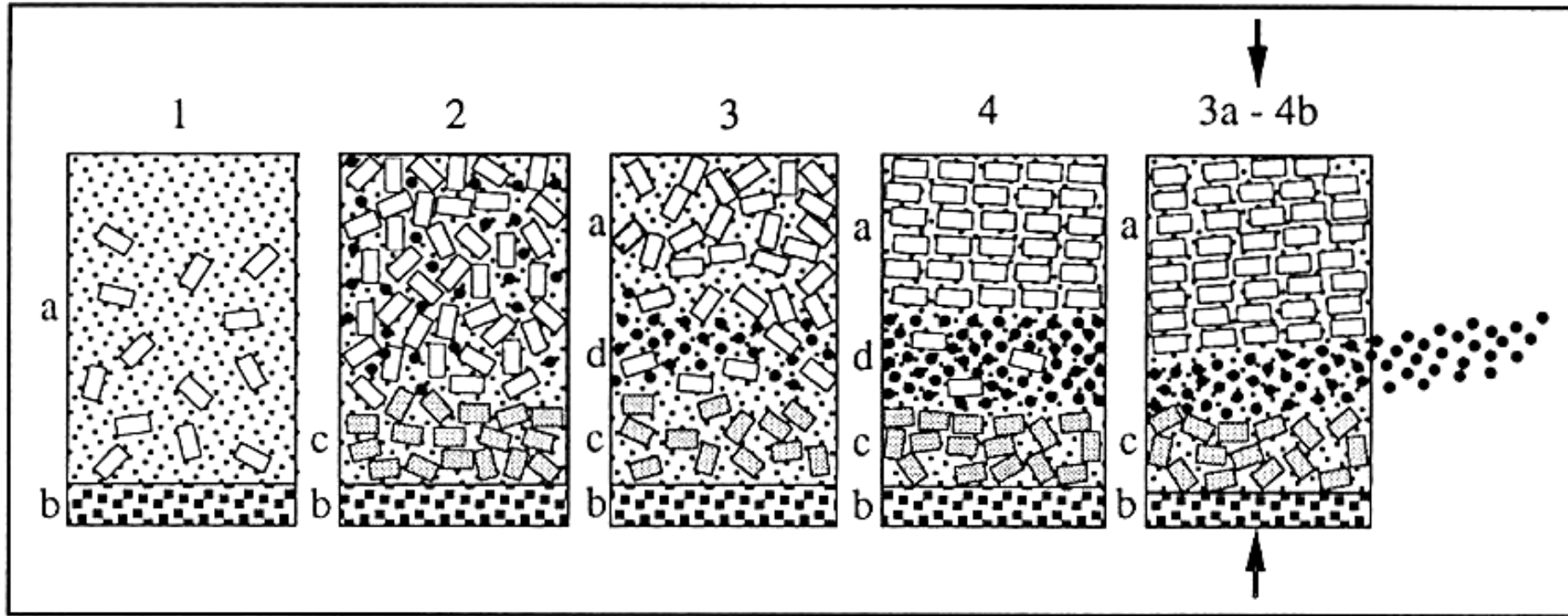


A. Grains of three minerals settle at different rates and produce three rock types of different composition.

Fractional crystallization of olivine and fractional segregation of sulfide from a cooling magma



Formation of Fe-Ti oxide deposit by residual liquid segregation and injection





Layered Chromite



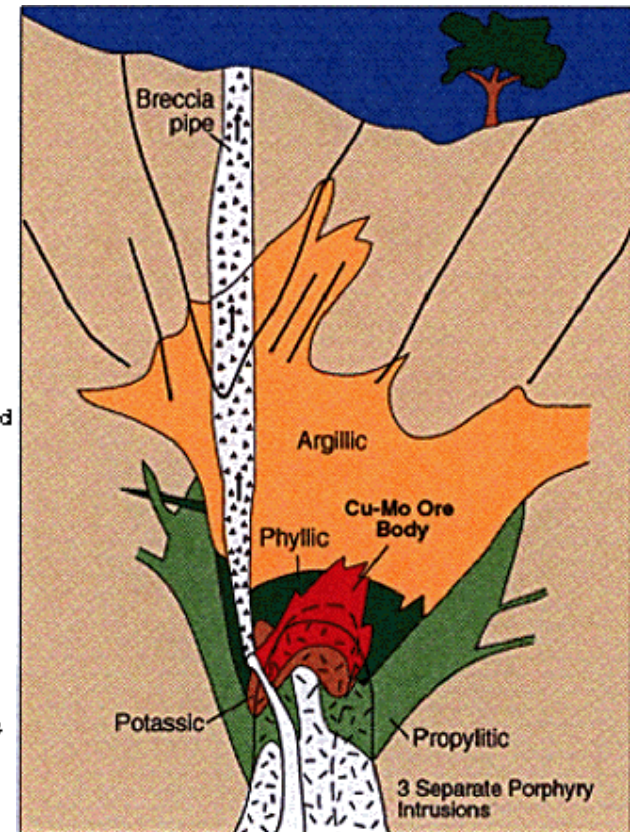
Platinum Group Elements

- 1. Ore minerals of PGE occur as inclusions in ferromagnesian minerals and chromites**
- 2. Due to siderophile character form tellurides, bismuthides, antimonides, tellurides and arsenides**
- 3. Due to chalcophile character form chalcogenides**
- 4. Due to affinity to chalcophile elements occur along with them as sulphides (Ni,Cu,Co,S)**

Major PGE producing / potential Regions in the world

1. Bushveld Complex, S.Africa
2. Stillwater Complex, Montana, USA
3. The Great Dyke, Zimbabwe
4. Norilisk-Talnakah,USSR
5. Sudbury, Canada
6. Jin Chuan,NW China
7. Munni Munni Complex, NW Australia

- Thermal gradients induce convection of water – leaching, redox rxns, and cooling create economic mineralization



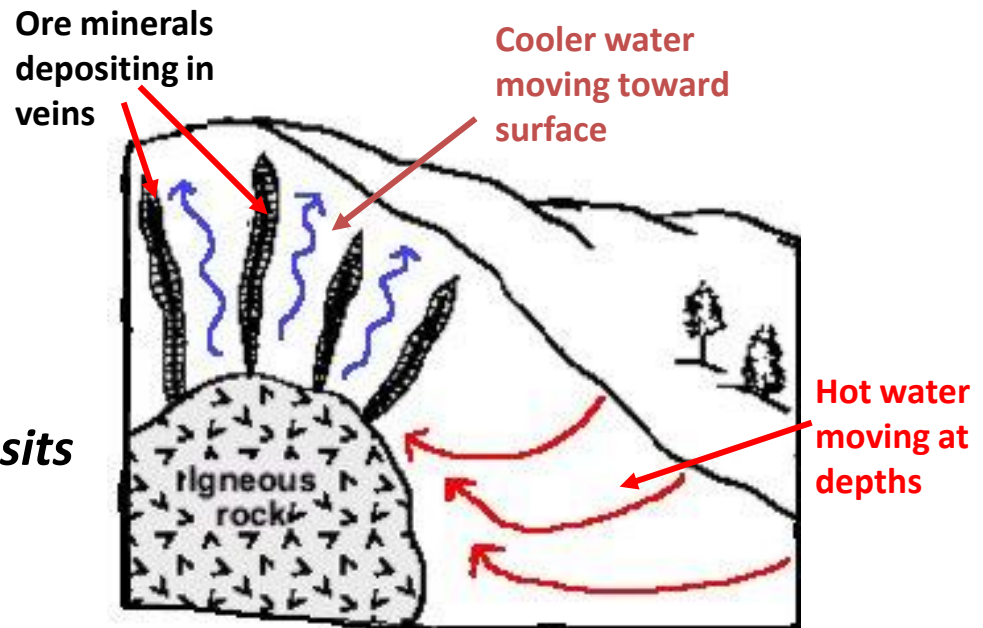
Hydrothermal Mineral Deposits

Minerals are concentrated by hot fluids flowing through fractures and pore spaces in rocks.

Hydrothermal deposits are produced when groundwater circulates down to depths and heats up, either by coming near a hot igneous body or by circulating to great depths which naturally heat the water because of the geothermal gradient.

Examples:

- 1) *Massive sulfide deposits*
- 2) *Vein deposits*
- 3) *Stratabound mineral deposits*



- In their journey through the rocks, these solutions may lose their mineral content by deposition in various kinds of openings in rocks to form-

Cavity-filling Deposits.

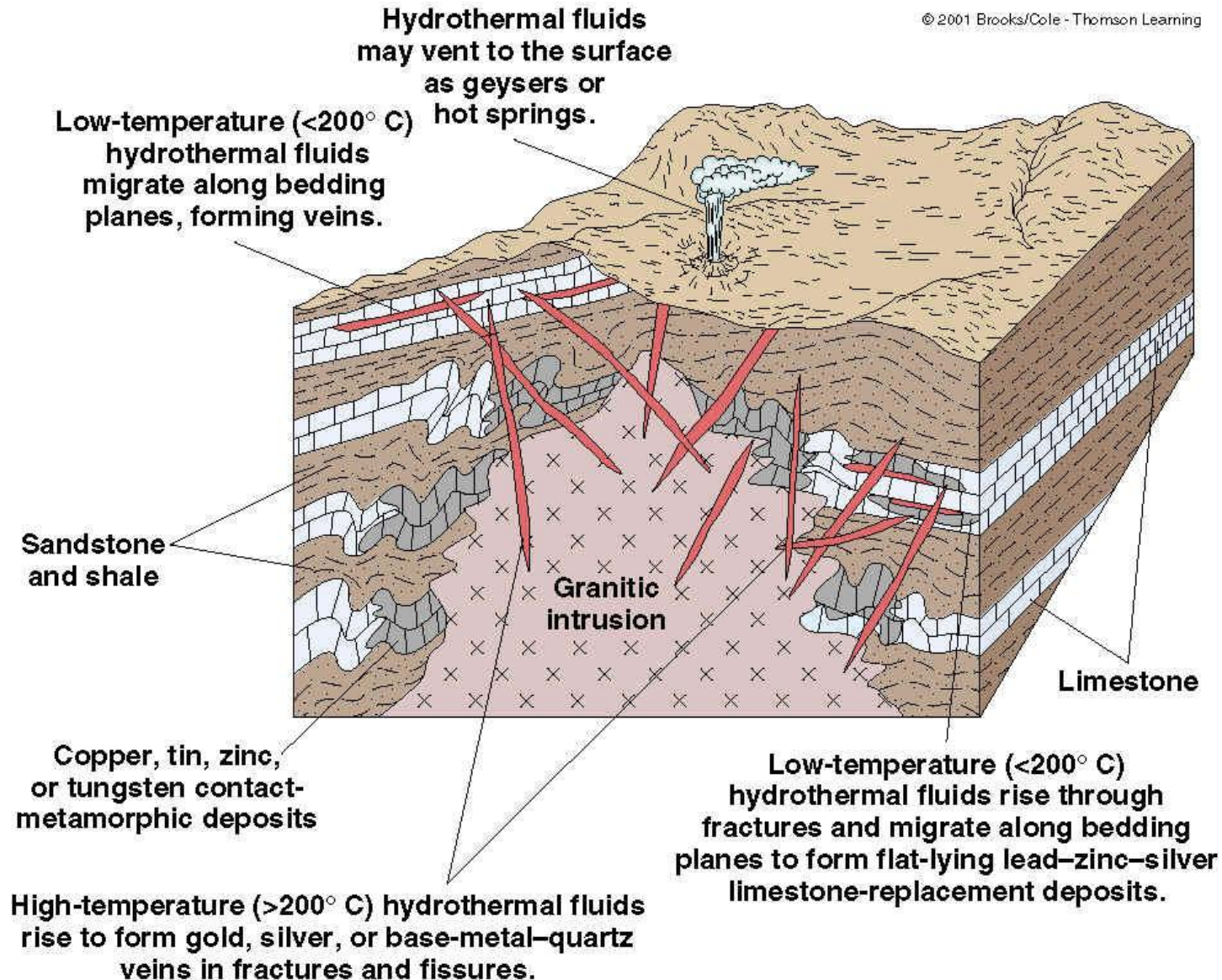
or by metasomatic replacement to form

Replacement Deposits.

Gradation between these may also occur.

In high temperatures Replacement Deposits dominate, whereas in low temperatures Cavity-filling dominates.

High temperature deposits are formed nearest the intrusive, intermediate at some distance & low temperature at farther outward. Accordingly named- Hypothermal, Mesothermal, & Epithermal.



Various kinds of openings in rocks

1. **Pore-spaces** :These are interstitial spaces between grains, capable of absorbing fluids.
2. **Permeability** :It is a measurement of yield of rocks & depends particularly upon the interconnection of Pore-spaces.
3. **Bedding planes** : These permits the ingress of Hydrothermal Solutions & replacement by ore of adjacent walls.
4. **Vesicles or “blow-holes”** : These are openings produced by expanding vapours, typical of the upper part of Basaltic lava flows.
5. **Fissures** :These are formed by compressive, tensile or torsional forces & are continuous tabular openings in rocks of considerable length & depth.
6. **Shear zone cavities** :
These results where fractures are expressed in innumerable closely spaced & more or less parallel discontinuous breaks.
7. **Folding & Warping** :
These are openings at fold-crests.
8. **Rock alteration opening** : Altered rocks become more porous & permits the ingress of mineralizing solutions. E.g. dolomitization of limestone makes it greater porous & permeable.



Fig. D.2.8 Different types of fissure veins. a. Sheeted veins, b. Linked veins, C. Chambered Dilation veins, d. en echelon, e. Branched veins.

Hydrothermal deposits

(A). Cavity-filling:

Ore solutions when deposited in cavity exhibit method & order.

Basically Temperature dependent order of crystallization from wall towards the centre of cavity i.e. vug.

1. Vein-filling where nearby sub-parallel crystals (usually Quartz) have grown almost at right angles to wall cavity constitutes **Comb-structure**.

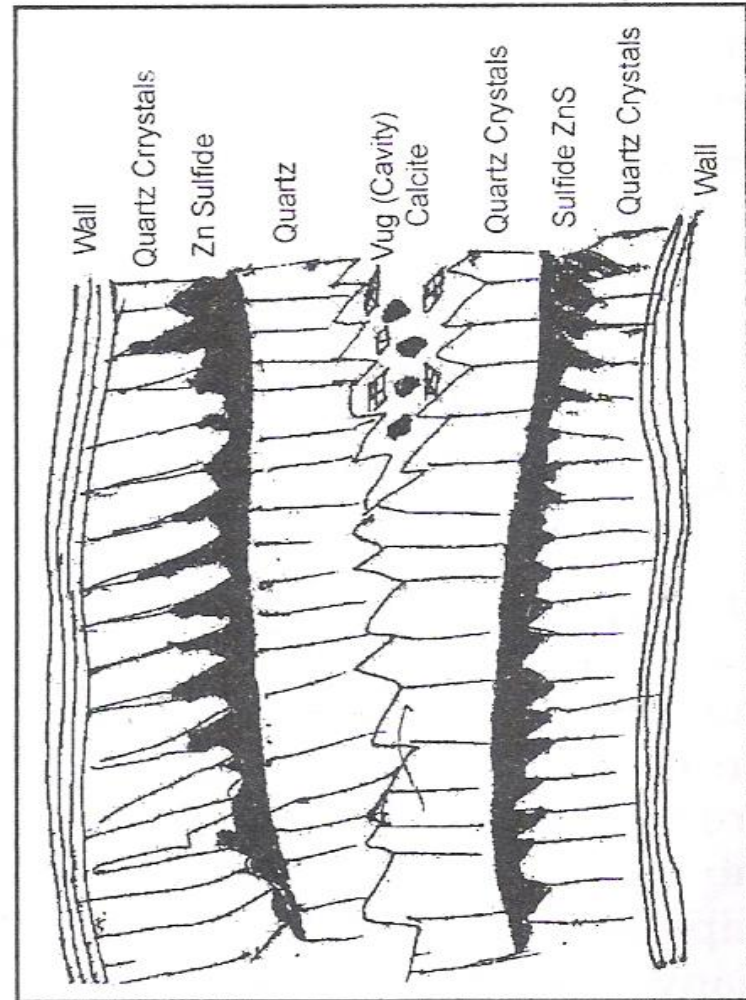


Fig. D.2.5 A comb Structure

Resulting mineral deposits

1. Fissure veins : Tabular or sheet like in shape, generally discordant opening in rock.
2. Shear-Zone deposits: Represents zone of ductile deformation between two rather undeformed rock masses that has connected, thin sheet like openings. E.g. Singhbhum shear zone(Jharkhand).
3. Stockworks : Consists of closely spaced interlacing networks of veinlets that are filled with ore minerals usually in various disseminated forms.
4. Saddle-reef : This ore body takes a shape of a saddle top triangle in the hinge zone of a tight fold. E.g. Gold fields of Hutti (Karnataka).
5. Ladder veins : This takes the form of regularly spaced, transverse, cracks & fractures. E.g. Aesbestos & magnesite deposits of Karnataka.
5. Breccia filling Deposits.
6. Solution Cavity filling deposits.
7. Pore space fillings.
- 8.Vesicular fillings.

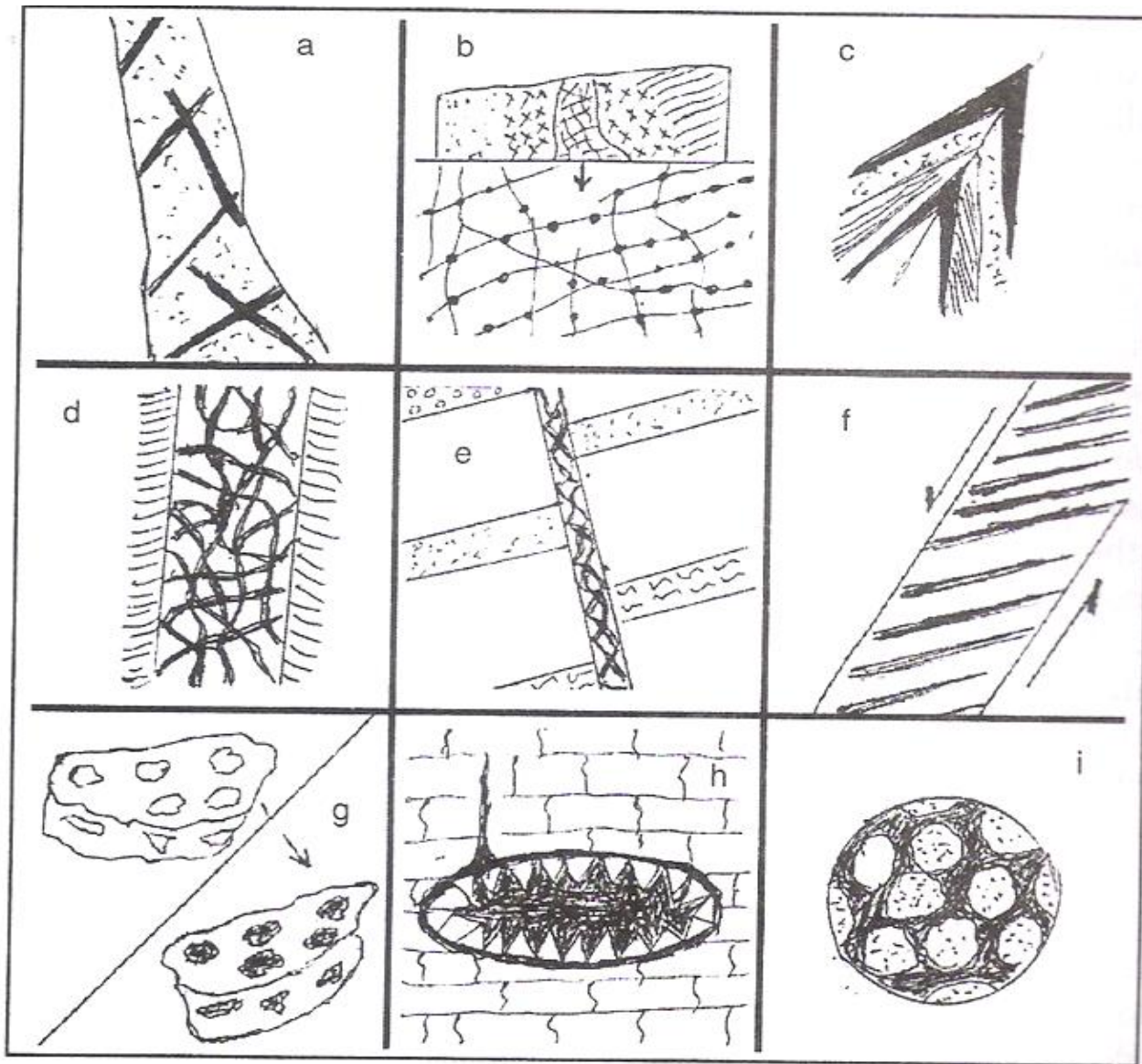
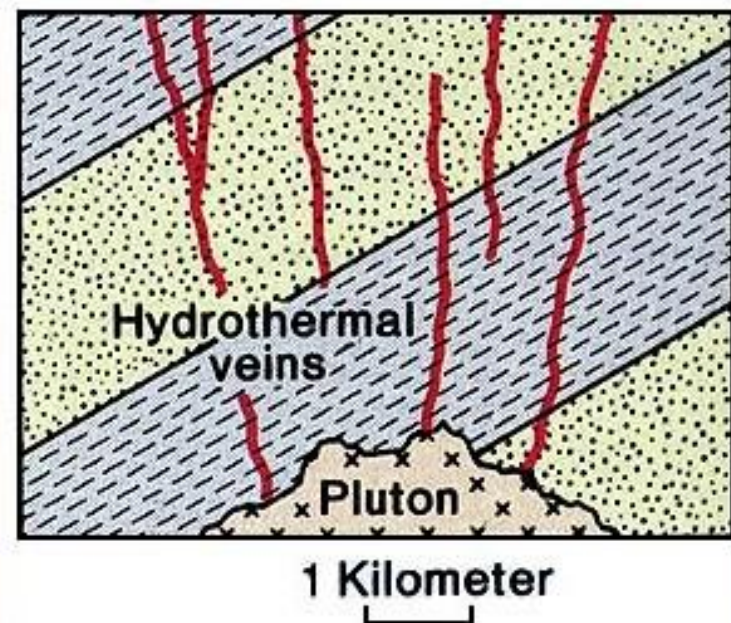


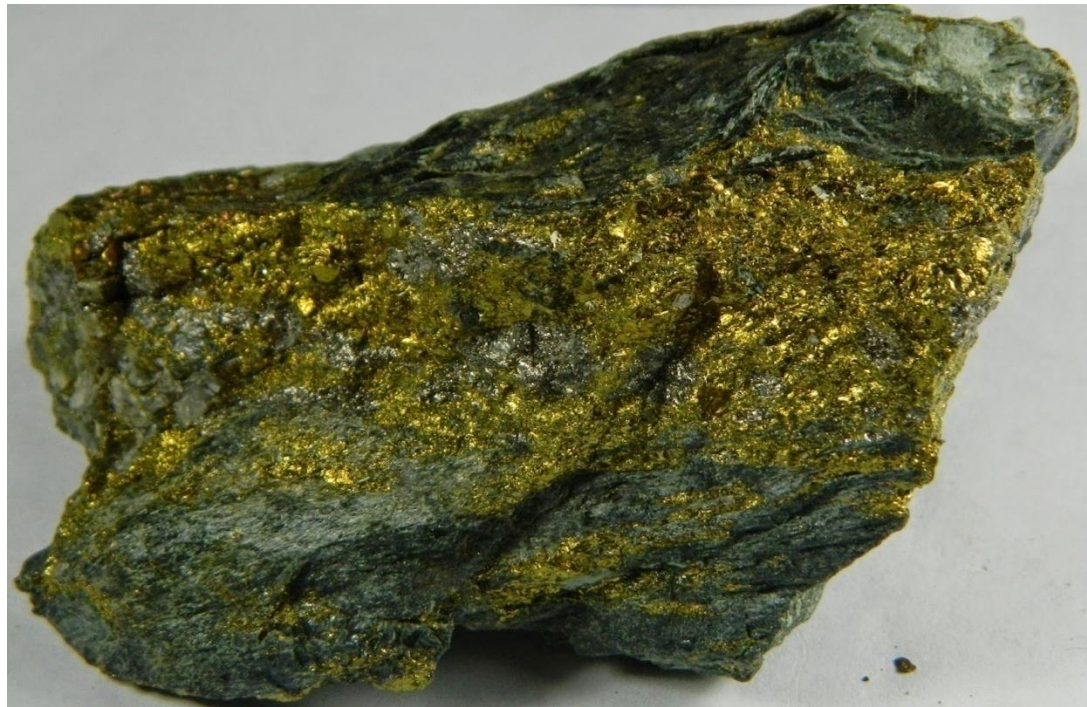
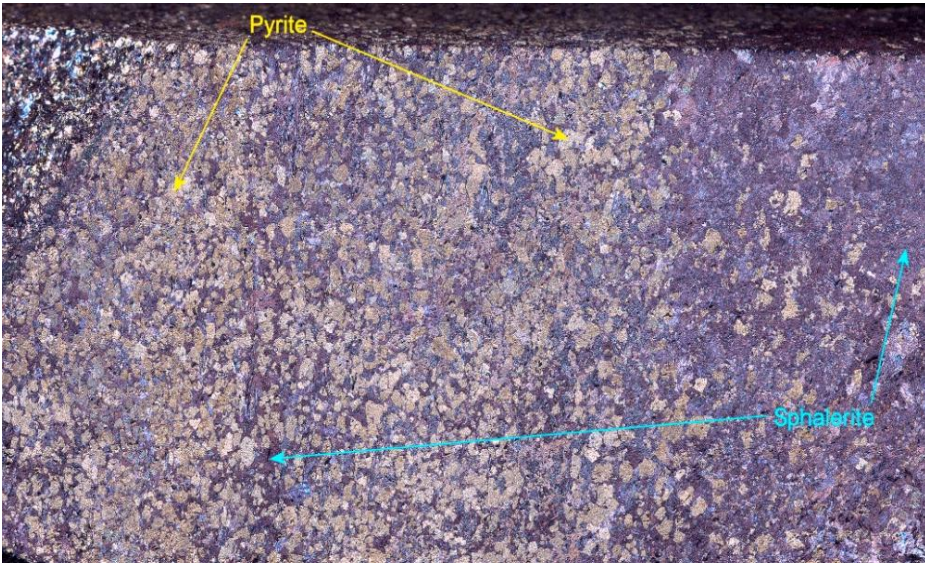
Fig. D.2.7 Different types of Cavity Fillings a. Ladder veins, b. Stock work, c. Saddle Reef, d. Pipe breccia, e. Fault breccia, f. Tension Cracks, g. Vesicular, h. Solution Cavity in limestone, i. Pore fillings.



Hydrothermal Ore Deposits

Figure 21.14 from: McGeary, David and Charles C. Plummer (1998) Physical Geology: Earth Revealed, 3rd ed. WCB/McGraw-Hill, p. 486.

Massive Sulfide Ores



Sedimentary Mineral Deposits

Minerals are concentrated by chemical precipitation from lake or sea water.

These mineral deposits form as a result of chemical sedimentation, where minerals are precipitated directly out of water.

Evaporite Deposits - Evaporation of lake water or sea water results in the loss of water and thus concentrates dissolved minerals in the remaining water. When the water becomes saturated with dissolved minerals, they precipitate from the water. Deposits of **halite** (table salt) and **gypsum** (used in plaster and wall board), result from this process.

Iron Formations - These deposits are of iron rich minerals that were deposited in marine environments during the Proterozoic. These formed as a result of iron-rich waters reacting with oxygen released by algae in the early stages of the evolution of life. The composition of sea water must have been drastically different than it is today.



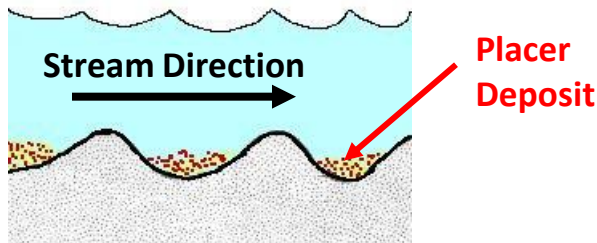
Placer Mineral Deposits

Minerals are concentrated by flowing surface waters depositing high density minerals either in streams or along coastlines.

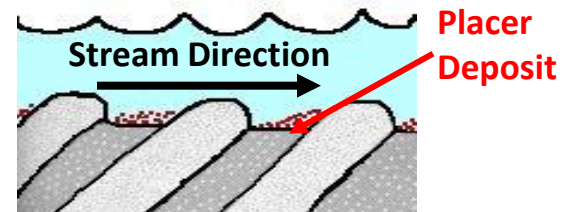
When the velocity of the water slows, minerals with a higher density are deposited. Heavy minerals like gold, diamond, and magnetite will be concentrated in areas where water current velocity is low. The lighter minerals (quartz) are carried away.

Gold originally form in hydrothermal veins, is eroded out of the veins and carried in streams where it was deposited in placer deposits.

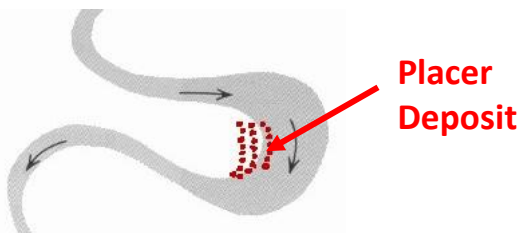
1) between ripple marks



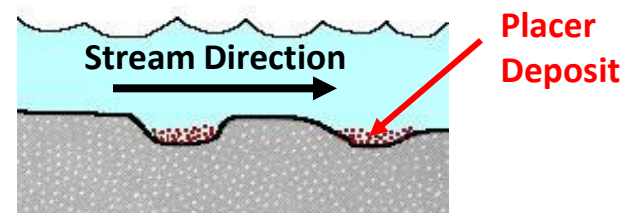
2) behind rock bars



3) on the inside of meandering streams



4) in holes on the bottom of a stream



Origin of Mineral Deposits

5) Residual Mineral Deposits

- Minerals are concentrated by chemical weathering processes.
- These deposits often form as a result of chemical weathering in warm tropical climates that receive high temperatures and high amounts of rainfall which produces highly leached soils rich in both iron and aluminium. Chemical weathering tends to remove the soluble materials, leaving the less soluble residues.
- Two common mineral deposits formed in this way are iron-rich **Limonite** and aluminium-rich **Bauxite**. Bauxite is the world's primary source of aluminium.

Residual Mineral Deposits

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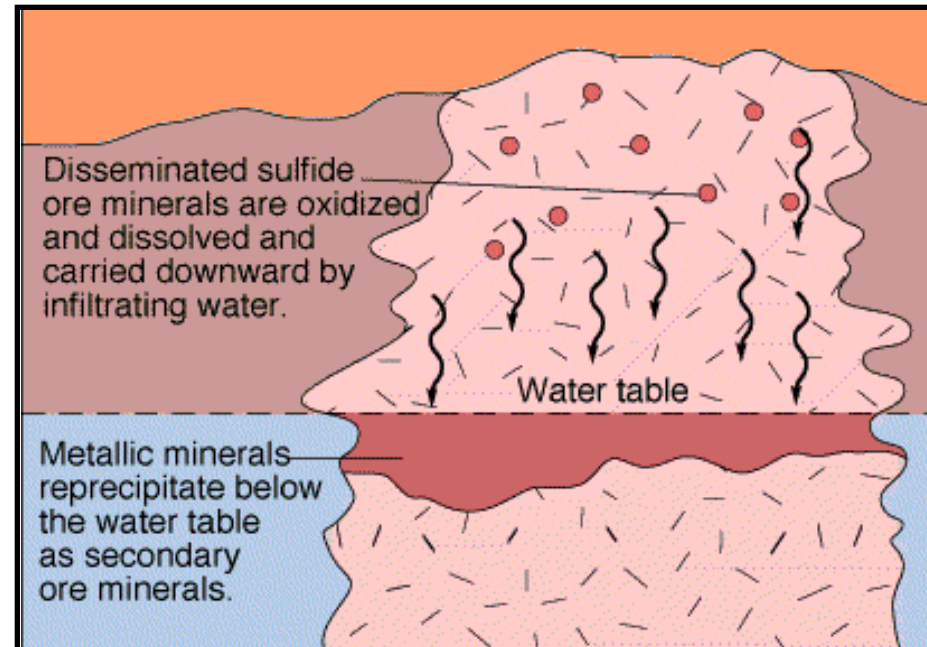
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Bauxite is the world's primary source of aluminium.

In addition, an existing mineral deposit can be turned in to a more highly concentrated mineral deposit by weathering in a process called secondary enrichment.



Grade & Tenor

The **lowest admissible limit of metallic** content of an ore mineral is known as its tenor. The metal content of an ore is called the tenor or grade, which is generally expressed as %, ppm, in case of precious metals, ounces per ton.

Grade represents the concentration of metal/metals in an ore body which depends upon the chemical composition and nature of ore mineral.

Tenor and Grade Concept for Fe Ore

- 1 cubic metre of pure hematite (Fe_2O_3) gives nearly 69.65% of Fe
- 1 cubic metre of quartzite with a total volume of 20% of hematite
- Grade will be 69.65/5

Tenor of Chalcopyrite

Chalcopyrite (CuFeS_2):

Atomic mass of Cu=63.546

Atomic mass of Fe=55.845

Atomic mass of S=32

Molecular Weight of Chalcopyrite

$=63.546+55.845+64$

$=183.39$

$\text{Cu} = 63.546 / 183.39 * 100 = 34.65$

Cut off grade-A cut-off grade is a grade below which the value of contained metal/mineral in a volume of rock does not meet certain specified economic requirements. The material above the lowest grade will be included in the potentially economic part of the deposit.

$$Ct = \frac{Pc}{Vm} 100$$

Resource and Reserve

Total Mineral Resource – A total mineral resource is a naturally occurring concentration of mineral raw material in or on the earth's crust of economic interest and with specified geological certainty.

Total Mineral Resource = Reserve + Remaining or additional resource

Mineral Reserve

Mineral Reserve- A mineral reserve is the economically mineable part of total Mineral resource (Measured or Indicated Mineral Resource.) as demonstrated by mineability assessments. In accordance with the stage of mineability assessment. Reserves are classified into the following two broad categories:

Proved Mineral Reserve- Proved Mineral Reserves are the economically mineable part of a Measured Mineral Resource. Demonstrated to be economically mineable by a feasibility study or actual mining activity. (111)

Probable Mineral Reserve- Probable Mineral Reserves are the economically mineable part of Indicated Mineral Resource and in some circumstances, Measured Mineral Resources. Demonstrated to be economically mineable by a Prefeasibility Study. (121) (122). A Probable Mineral Reserve has a lower level of confidence than a Proved Mineral Reserve

MINERAL RESOURCES

A Mineral Resource is a concentration or occurrence of material of intrinsic economic interest in or on the Earth's crust in such form and quantity that there are reasonable prospects for eventual economic extraction.

- **Inferred Mineral Resources:** Part of a Mineral Resource for which tonnage, grade and mineral content can be estimated based on geological evidence and assumed, but not verified, geological and/or grade continuity.
- **Indicated Mineral Resources:** Part of a Mineral Resource for which tonnage, densities, shape, physical characteristics, grade and mineral content can be estimated with a reasonable level of confidence.
- **Measured Mineral Resources:** A Measured Mineral Resource is estimated in the same manner as an Indicated Mineral Resource, except that (1) it can be estimated with a high level of confidence; (2) the information must be detailed and reliable; and (3) the locations are spaced closely enough to confirm geological and/or grade continuity

Exploration Results

MINERAL RESOURCES

MINERAL RESERVES

Inferred

Indicated

Measured

Probable

Proved

Increasing level of
geological
knowledge and
confidence

Consideration of mining, processing, metallurgical, economic,
marketing, legal, environmental, infrastructure, social,
and governmental factors
(the "Modifying Factors").

Economic Minerals and Associated Rocks

Igneous Rocks:

Norite (Ultramafic rock) : Ni, Cu, PGE

Diabase : Ag, Co, Ni.

Diorite-Monzonite, Granite: Cu

Granodiorite : Porphyry copper

Quartz Monzonite : Mo

Granitoids : Au, Cu, Sn, W, Li, Nu, Ba, Su, Bi, Mo

Basic volcanics : Massive sulphides Cu-Zn-Pb